

The Helmholtz Machine

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Intuition

From (Restricted) Boltzmann machine to Helmholtz Machine

Boltzmann machine's energy function (a column vector $\{0, 1\}$)

$$p(\mathbf{x}) = \frac{\exp(-E(\mathbf{x}))}{Z}$$

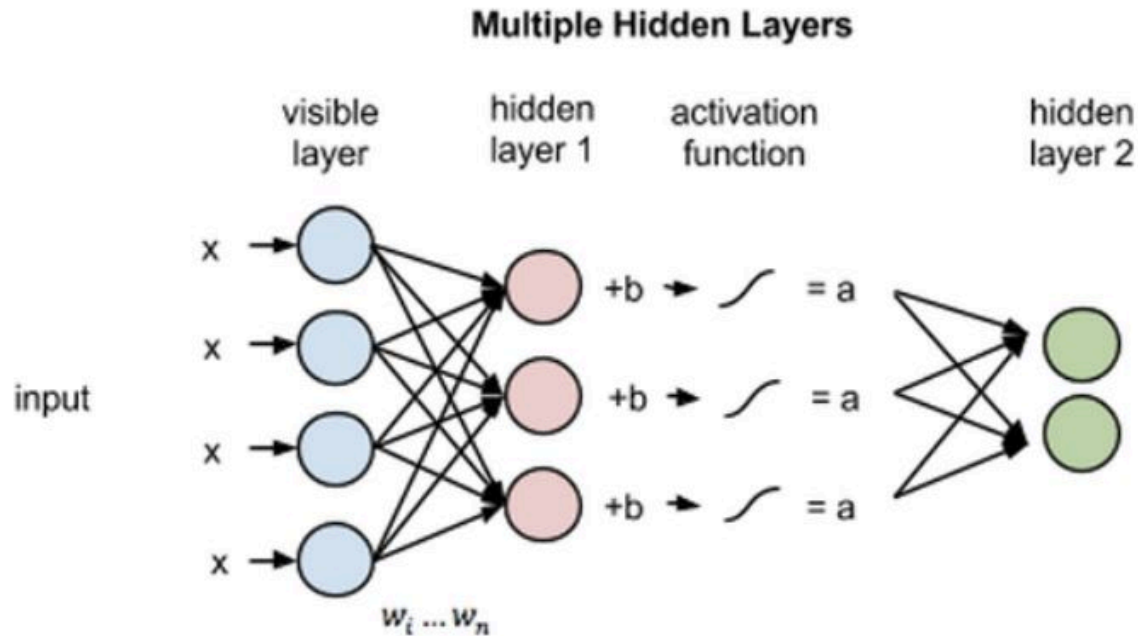
$$E(\mathbf{x}) = -\mathbf{x}^T U \mathbf{x} - \mathbf{b}^T \mathbf{x} = -\sum_{i,j} w_{i,j} x_i x_j - \sum_i w_i x_i$$

Input can be divided into two parts:

$$\mathbf{x} = (\mathbf{v}, \mathbf{h})$$

Restricted Boltzmann Machine

visible variables and hidden variables are fully connected



Restricted Boltzmann Machine

$$p(\mathbf{v}, \mathbf{h}) = \frac{\exp(-E(\mathbf{v}, \mathbf{h}))}{Z}$$

$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{v}^T W \mathbf{h} - b^T \mathbf{v} - c^T \mathbf{h}$$

Then we can get the conditional distribution for hidden variables given visible variables (encoder):

$$p(\mathbf{h} | \mathbf{v}) = \prod_{j=1}^{n_h} \sigma\left((2h - 1) * \left(c + W^T \mathbf{v}\right)\right)_j$$

Restricted Boltzmann Machine

Decoder:

$$p(\mathbf{v} \mid \mathbf{h}) = \prod_{i=1}^{n_v} \sigma\left((2v_i - 1) * (b_i + W_i^T \mathbf{h})\right)$$

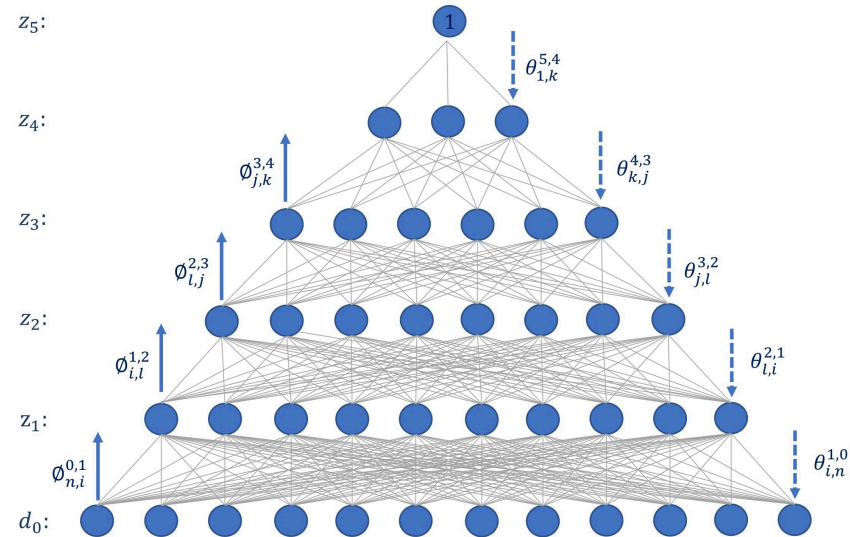
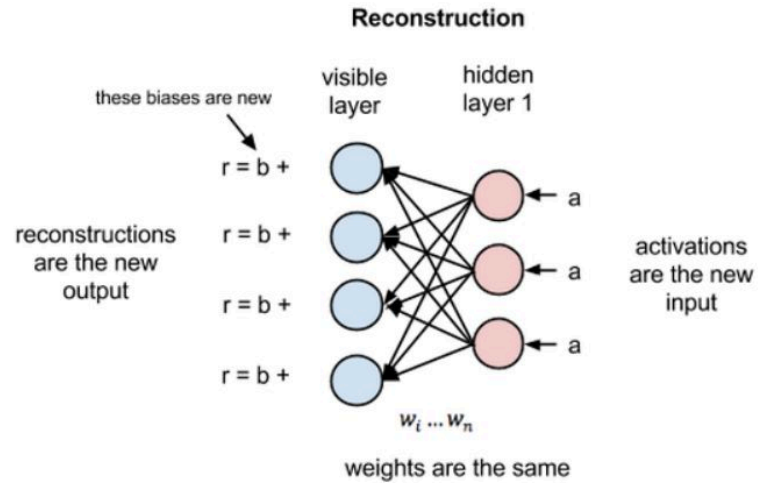
Learning with Contrastive Divergence.

Helmholtz Machine

Helmholtz Machine is a hierarchical full-connected feedback NN.

RBM can be viewed as a Helmholtz machine with only **one layer of hidden units** and an **implicit posterior distribution**.

Helmholtz Machine



Divide generative model and recognition model

Why it is called Helmholtz Machine? Two models are trained iteratively.

MLE, theta is model parameters:

$$\theta = \arg \max(\theta) \sum_{i=1}^n \log p(\mathbf{v}_i | \theta)$$

$$\theta = \arg \max(\theta) \sum_{i=1}^n \log p(\mathbf{v}_i, \mathbf{h} | \theta)$$

Real posterior distribution is intractable

$$p(\mathbf{h} \mid \mathbf{v}, \theta) = \frac{p(\mathbf{h}, \mathbf{v} \mid \theta)}{p(\mathbf{v} \mid \theta)}$$

Approximation with Q:

$$Q(\mathbf{h} \mid \mathbf{v}) = \prod_{i=1}^Q (h_i \mid \mathbf{v})$$

ELBO (Evidence Lower Bound) from variational inference:

$$\log p(\mathbf{v} \mid \theta) = E_Q[\log p(\mathbf{v}, \mathbf{h} \mid \theta) - \log Q(\mathbf{h} \mid \mathbf{v})] + D_{KL}(Q(\mathbf{h} \mid \mathbf{v}) \parallel p(\mathbf{h} \mid \mathbf{v}, \theta))$$

Real posterior distribution is intractable

$$D_{KL}(Q(\mathbf{h} \mid \mathbf{v}) \parallel p(\mathbf{h} \mid \mathbf{v}, \theta)) > 0$$

Therefore, the ELBO is a lower bound of the log-likelihood.

$$L(\theta, Q) = E_Q[\log p(\mathbf{v}, \mathbf{h} \mid \theta) - \log Q(\mathbf{h} \mid \mathbf{v})]$$

That is, $\log p(\mathbf{v} \mid \theta) \geq L(\theta, Q)$.

Introduce free energy

$$F(\mathbf{v}, \theta) = -E_Q[\log p(\mathbf{h}, \mathbf{v} \mid \theta)] - H(Q(\mathbf{h} \mid \mathbf{v}))$$

The objective is to minimize the free energy.

Now we have two models: $p(\mathbf{v}, \mathbf{h} \mid \theta)$ and $Q(\mathbf{h} \mid \mathbf{v}; \varphi)$

We can use the free energy to optimize the parameters.

$$\theta = \arg \min(\theta) F(\mathbf{v}, \theta)$$

$$\varphi = \arg \min(\varphi) F(\mathbf{v}, \theta)$$

Bi-level optimization

Wake-sleep algorithm The wake phase is to minimize the free energy, and the sleep phase is to maximize the free energy.

Wake phase: target is minimize $F(\mathbf{v}, \theta)$, i.e., maximize $L(\theta, Q)$. use real data pass Q to get posterior distribution. then update θ .

Sleep phase: generate data with $p(\mathbf{v}, \mathbf{h} \mid \theta)$. pass generated data to Q to get posterior distribution. then update ϕ .

Modern development

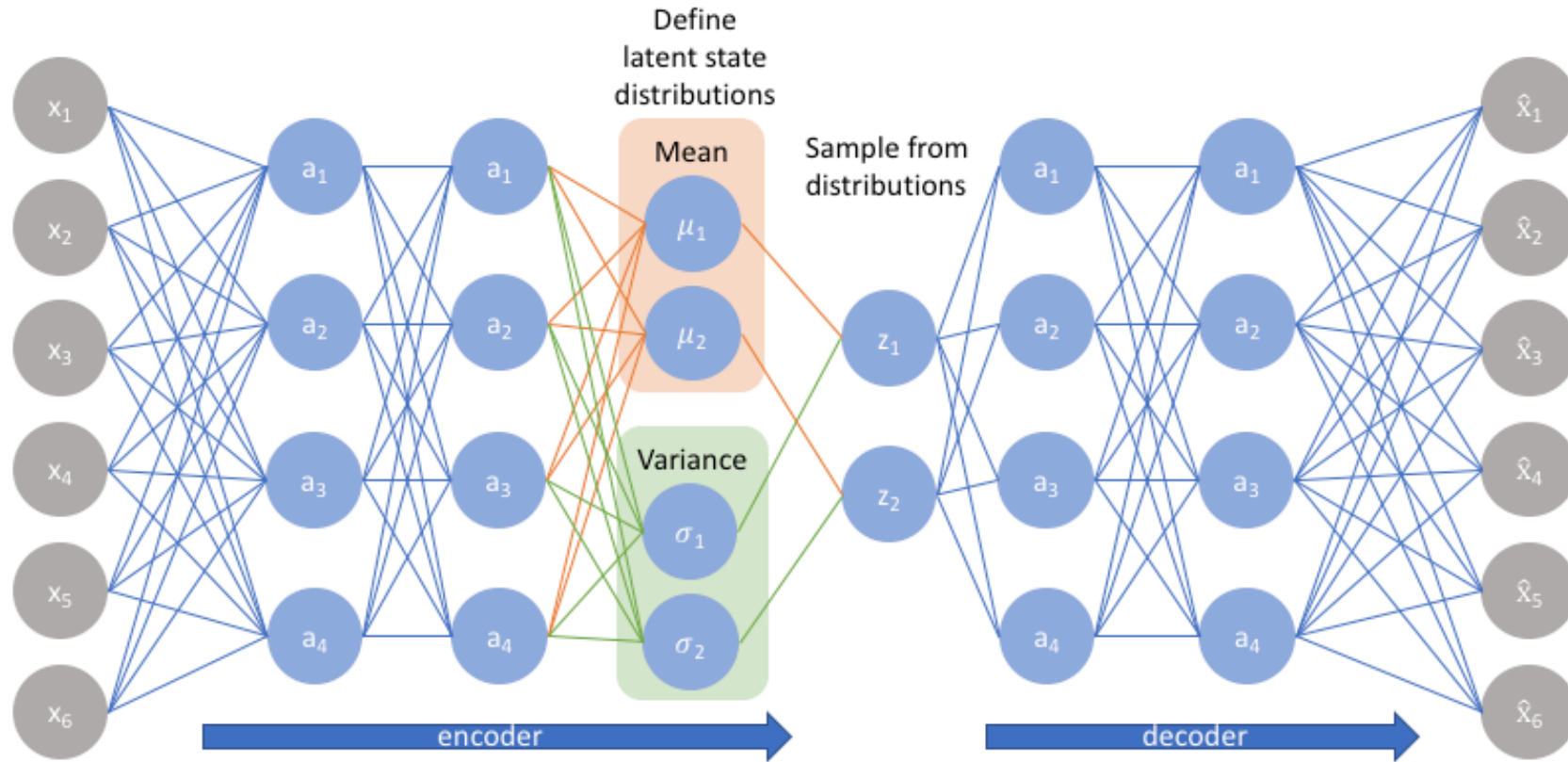
Variational Autoencoders (VAEs)

Generative Adversarial Networks (GANs)

Flow-based models

Diffusion models

Variational Autoencoders (VAEs)



Works on LLM

KV-cache compression via importance evaluation

Block Attention